

5 Fig. 5.1 shows the area of land that was used to grow genetically modified (GM) crops in the USA, Brazil, India and China from 2004 to 2015.

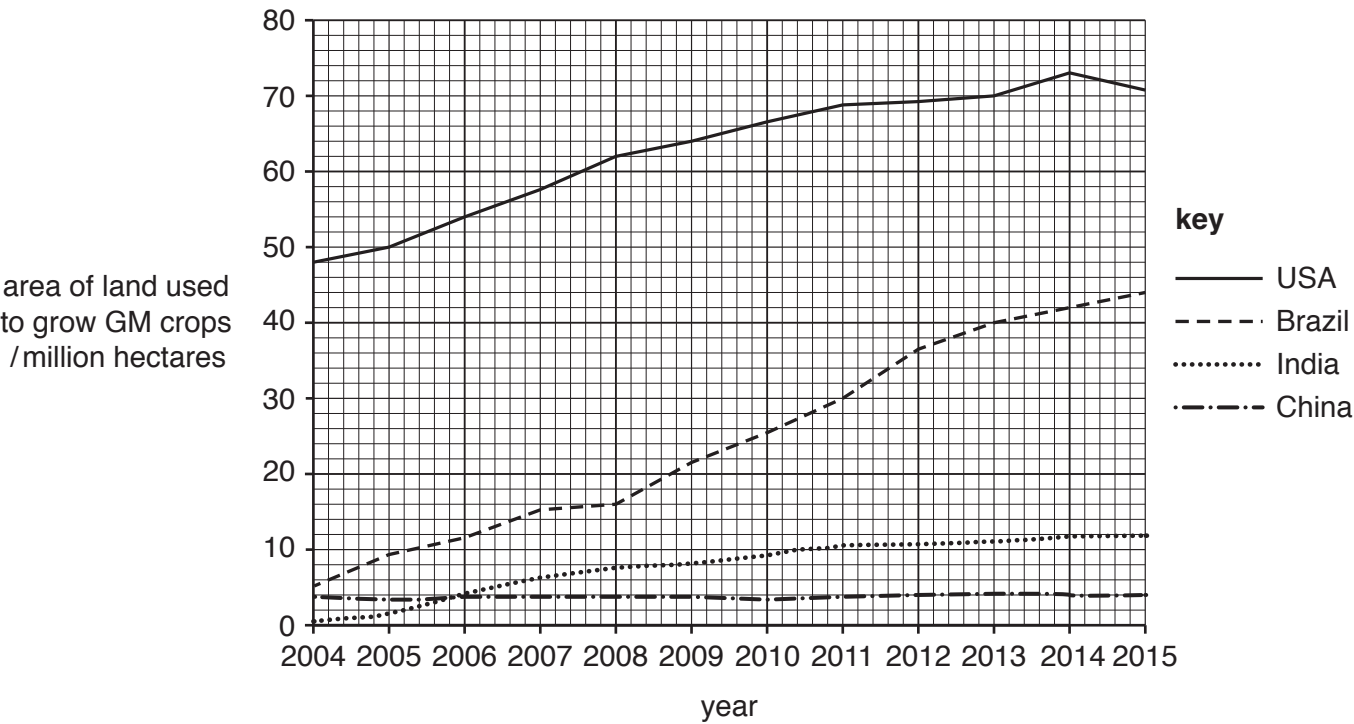


Fig. 5.1

(a) (i) Suggest reasons why the area of land used to grow GM crops in the USA is greater than the area of land used to grow GM crops in Brazil.

.....

.....

.....

..... [2]

(ii) Describe the differences in the area of land that was used to grow GM crops in China and the area of land that was used to grow GM crops in India, over the time shown in Fig. 5.1.

.....

.....

.....

.....

.....

.....

..... [3]

- (b) (i) Genetic modifications in crops can provide resistance to insect pests. Bt maize is one type of GM crop that has been produced to be resistant to insect pests.

State **two** benefits to farmers of insect resistance in crops.

.....

.....

.....

..... [2]

- (ii) Describe the difference between Bt maize and non-GM maize that explains why Bt maize is resistant to insects.

.....

.....

.....

..... [2]

- (c) (i) State **two** reasons why people may have objections to the growth of insect-resistant GM crops.

.....

.....

.....

..... [2]

- (ii) In 2015, the percentage of the USA population that objected to the use of GM crops was lower than in 2005.

Suggest why a smaller percentage of the USA population objected to the use of GM crops in 2015.

.....

..... [1]

[Total: 12]